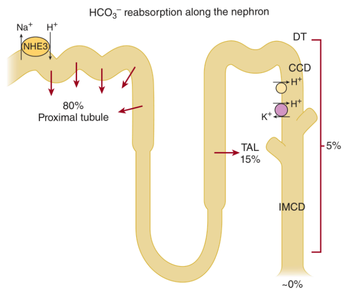



**Fig. 15.1 $HCO_3^-$ reabsorption along the nephron.** The proximal tubule reabsorbs approximately 80% of the filtered $HCO_3^-$. Substantial amounts are also reabsorbed in the thick ascending limb. The remaining amounts (~5%) are reabsorbed in the collecting duct, which is critical for final regulation of acid excretion. The mechanisms are described in Chapter 9, but the main apical transporters (Na-H exchange and $H^+$-ATPase) are illustrated. Little $HCO_3^-$ (~0%) remains in the final urine.

Fig_15_1

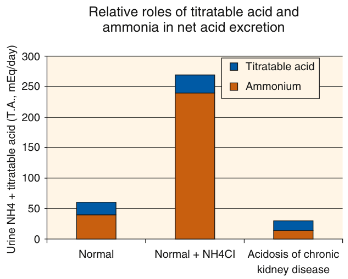



**Fig. 15.2 Net acid excretion with typical Western diets, acid loading, and acidosis in chronic kidney disease (CKD).** Ammonium excretion has the capability for the most increase in acidosis and is the component most compromised in CKD. *T.A.*, Urine titratable acidity. (From Gauthier P, Simon EE, Lemann J Jr. Acidosis of chronic renal failure. In: DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders*. Philadelphia: Saunders; 2002. pp. 207–216.)

Fig_15_2

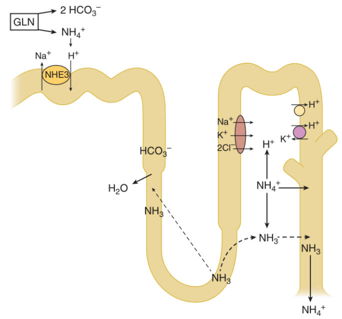



**Fig. 15.3 Synchrony of regulation of ammonium production (from glutamine [GLN] precursors, and excretion).** Process allows generation of "new" $HCO_3^-$ by the kidney. $NH_4^+$ excretion is regulated in response to changes in systemic acid-base and $K^+$ balance. Contributing segments include the proximal convoluted tubule, proximal straight tubule, thin descending limb, thick ascending limb, and medullary collecting duct. Upregulated by acidosis and hypokalemia. Inhibited by hyperkalemia.

Fig_15_3

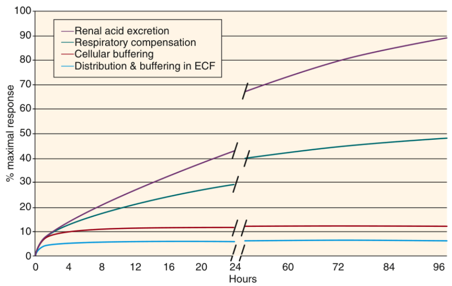



**Fig. 15.4 Conceptual illustration of time course of acid-base compensatory mechanisms in response to a metabolic acid load.** Component processes include rapid distribution and extracellular buffering mechanisms and cellular buffering events. But these most rapid mechanisms have limited capacity. Slower are respiratory and renal regulatory processes. *ECF*, Extracellular fluid.

Fig_15_4

**Fig. 15.5 Acid-base nomogram (map).** *Shaded areas* represent the 95% confidence limits of the normal respiratory and metabolic compensations for primary acid-base disturbances. Data falling *outside the shaded areas* denote a mixed disorder if a laboratory error is not present (see text).

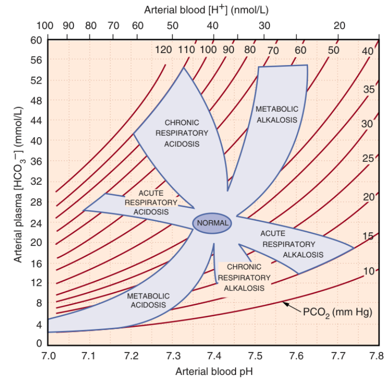



Fig_15_5

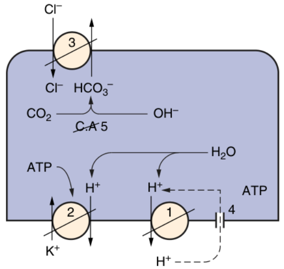



**Fig. 15.6 Type A intercalated cell of the collecting duct displaying five pathophysiologic defects that could result in classical distal renal tubular acidosis: defective $H^+$-ATPase, defective $H^+$-$K^+$-ATPase, defective $HCO_3^-/Cl^-$ exchanger, $H^+$ leak pathway, and defective intracellular carbonic anhydrase (type II).** *ATP*, Adenosine triphosphate.

Fig_15_6

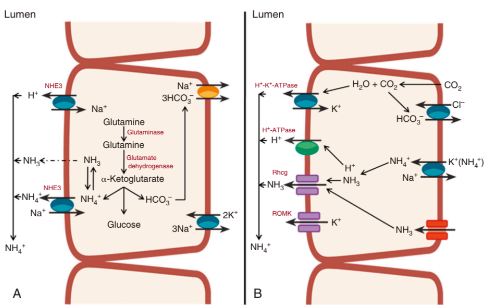



**Fig. 15.7 Cell models of ammonia synthesis and excretion pathways.** (A) Proximal convoluted tubule. Ammonia is derived from glutamine precursors to produce two $NH_4^+$ and two $HCO_3^-$ molecules through an enzymatic pathway activated by acidemia and hypokalemia and inhibited by alkalemia and hyperkalemia. (B) Type A intercalated cell in collecting tubule. Ammonium entry across basolateral membrane through substitution of $NH_4^+$ for $K^+$ on $K^+$ conductance and secreted across apical membrane via ROMK or RhCG (see text). In both A and B, $NH_3$ diffusion coupled with $H^+$ secretion traps $NH_4^+$ in the tubule lumen.

Fig_15_7

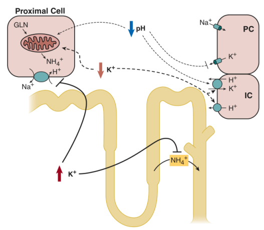



**Fig. 15.8 Diagram of some of the renal interactions of acid-base status and $K^+$.** Proximal tubule cell, a collecting duct principal cell (PC), and intercalated cell (IC) are shown. Acidosis (↓ pH) causes renal potassium retention by several actions including inhibiting secretory $K^+$ channels, stimulating $H^+$-$K^+$-ATPase, which reabsorbs $K^+$, and increasing ammoniagenesis, which may have secondary actions that block $K^+$ secretion. Alkalosis can have the reverse effects. The effects also work in the other direction: Potassium depletion (↓ $K^+$) stimulates ammoniagenesis and $H^+$ secretion/bicarbonate reabsorption, as well as distal acid secretion, all of which may cause metabolic alkalosis. Hyperkalemia can have some of the opposite effects, such as inhibiting thick ascending limb ammonium transport and inhibiting urinary ammonium excretion, an effect that may cause acidosis. (From Hamm LL, Hering-Smith KS, Nakhoul NL. Acid-base and potassium homeostasis. *Semin Nephrol.* 2013;333:257–264.)

Fig_15_8

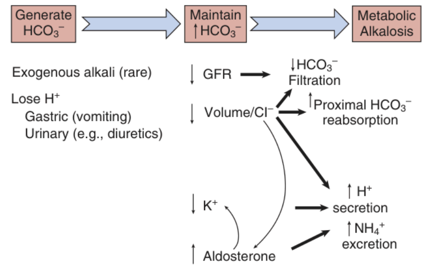
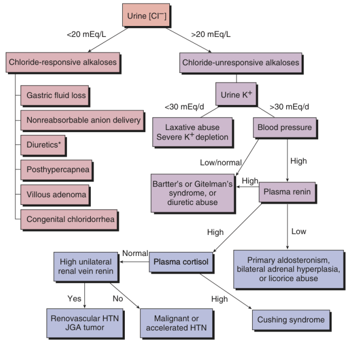



**Fig. 15.9 Pathophysiologic basis of the generation and maintenance of chronic metabolic alkalosis.** Paradoxical stimulation of bicarbonate absorption ($H^+$ secretion) and $NH_4^+$ production and excretion is the combined result of $Cl^-$ deficiency (with reduction in GFR), $K^+$ deficiency, and secondary hyperaldosteronism. *GFR*, Glomerular filtration rate.

Fig_15_9




* After diuretic therapy

**Fig. 15.10 Diagnostic algorithm for metabolic alkalosis, based on the spot urine $Cl^-$ and $K^+$ concentrations.** *HTN,* Hypertension; *JGA,* juxtaglomerular apparatus.

Fig_15_10

**Table 15.1 Acid-Base Abnormalities and Appropriate Compensatory Responses for Simple Disorders**

| Primary Acid-Base Disorders | Primary Defect | Effect on pH | Compensatory Response | Expected Range of Compensation | Limits of Compensation |
|---|---|---|---|---|---|
| Respiratory acidosis | ↑ $PCO_2^-$<br>Alveolar hypoventilation | ↓ | $HCO_3^-$ ↑<br>Chronic: ↑ Renal reabsorption & production $HCO_3^-$ | Acute<br>Δ $[HCO_3^-]$ = +1 mEq/L for each ↑ Δ$PCO_2$ of 10 mm Hg<br>Chronic<br>Δ $[HCO_3^-]$ = +4 mEq/L for each ↑ Δ$PCO_2$ of 10 mm Hg | Acute<br>$[HCO_3^-]$ = 38 mEq/L<br>Chronic<br>$[HCO_3^-]$ = 45 mEq/L |
| Respiratory alkalosis | ↓ $PCO_2$<br>Alveolar hyperventilation | ↑ | $HCO_3^-$ ↓<br>Chronic: ↓ Renal $HCO_3^-$ reabsorption | Acute<br>Δ $[HCO_3^-]$ = -2 mEq/L for each<br>↓ Δ$PCO_2$ of 10 mm Hg<br>Chronic<br>Δ $[HCO_3^-]$ = -5 mEq/L for each<br>↓ Δ$PCO_2$ of 10 mm Hg | Acute<br>$[HCO_3^-]$ = 18 mEq/L<br>Chronic<br>$[HCO_3^-]$ = 15 mEq/L |
| Metabolic acidosis | ↓ $HCO_3^-$<br>Loss of $HCO_3^-$ or gain of $H^+$ | ↓ | ↓ $PCO_2$<br>Alveolar hyperventilation | $PCO_2 = 1.5[HCO_3^-] + 8 \pm 2$<br>$PCO_2$ = last 2 digits of pH x 100<br>$PCO_2 = 15 + [HCO_3^-]$ | $[PCO_2]$ = 15 mm Hg |
| Metabolic alkalosis | ↑ $HCO_3^-$<br>Gain of $HCO_3^-$ or loss of $H^+$ | ↑ | ↑ $PCO_2$<br>Alveolar hypoventilation | $PCO_2$ = +0.6 mm Hg for Δ $[HCO_3^-]$ of 1 mEq/L<br>$PCO_2 = 15 + [HCO_3^-]$ | $[PCO_2]$ = 55 mm Hg |

*$PCO_2$,* Carbon dioxide pressure.
Adapted from Bidani A, Tauzon DM, Heming TA. Regulation of whole body acid-base balance. In DuBose TD, Hamm LL, eds. *Acid-Base and Electrolyte Disorders: A Companion to Brenner and Rector's the Kidney*. Philadelphia: Saunders; 2002:1–2.

Table_15_1

**Table 15.2 Systematic Method for Diagnosis of Simple and Mixed Acid-Base Disorders**

1. Measure arterial blood gas and electrolyte concentrations simultaneously.
2. Determine whether the compensation is appropriate for a simple acid-base disorder; if inappropriate, a mixed acid-base disorder is present (see Table 15.1 and Fig. 15.5).
3. Calculate the anion gap (corrected for albumin) to determine the presence of a high AG metabolic acidosis: Appreciate the four major categories of high anion gap acidoses: ketoacidosis, lactic acidosis, renal failure acidosis, toxin- or poison-induced acidosis. Appreciate the two major causes of non–AG acidoses: gastrointestinal loss of $HCO_3^-$ and renal loss of $HCO_3^-$.
4. Compare the relative changes of $HCO_3^-$ and AG (Δ$HCO_3^-$ and ΔAG, respectively) to look for mixed disorders (see text).

Table_15_2

**Table 15.3 Causes of Abnormal Anion Gap**

**Increased Anion Gap**

Increased Anions (not $Cl^-$ or $HCO_3^-$)
↑ **Albumin**[a]
Alkalosis
↑ Inorganic anions
  Phosphate
  Sulfate
↑ **Organic anions**
  **L-Lactate**
  D-Lactate
  **Ketones**
  **Uremic**
↑ Exogenously supplied anions
  **Salicylate**
  Paraldehyde
  **Ethylene glycol**
  Propylene glycol
  **Methanol**
  Toluene
  Pyroglutamic acid (5-oxyprolene)
Other toxins
  **Uremic**
  Hyperosmolar, nonketotic states

**Myoglobinuric Acute Renal Failure**

**Decreased Cations (not $Na^+$)**

↓ $Ca^{2+}$, $Mg^{2+}$

**Decreased Anion Gap**

**Increased Cations (not $Na^+$)**

↑ $Ca^{2+}$, $Mg^{2+}$
↑ $Li^+$
↑ Immunoglobulin G

**Decreased Anions (not $Cl^-$ or $HCO_3^-$)**

**Hypoalbuminemia**[a]
Acidosis

**Laboratory Error**

Hyperviscosity
Bromism

[a]For each decline in albumin by 1 g/dL from normal (4.5 g/dL), the anion gap decreases by 2.5 mEq/L.
The items in bold are most common or notable.

Table_15_3

**Table 15.4 Clinical Causes of High Anion Gap and Normal Anion Gap Acidosis**

**High Anion Gap Acidosis**

Ketoacidosis
  **Diabetic ketoacidosis**
  **Alcoholic ketoacidosis**
  Starvation ketoacidosis
Lactic acidosis
  **L-Lactic acidosis (types A and B)**
  D-Lactic acidosis
**Renal failure acidosis**
Toxin-induced acidosis
  **Ethylene glycol**
  **Methyl alcohol**
  **Salicylate**
  Propylene glycol
  Pyroglutamic acid (5-oxyprolene)

**Nonanion Gap Acidosis**

Gastrointestinal loss of $HCO_3^-$ (negative urine anion gap)
  **Diarrhea**
  External fistula
Renal loss of $HCO_3^-$ or failure to excrete $NH4^+$
  Proximal renal tubular acidosis (RTA) (low serum $K^+$)
  Classical distal renal tubular acidosis (low serum $K^+$)
  **Generalized distal renal tubular defect (high serum $K^+$)**
  **Drugs that cause RTA**
  Carbonic anhydrase inhibitors (mixed proximal-distal RTA)
  Amphotericin B ("gradient" classical distal RTA)
Miscellaneous
  $NH_4Cl$ ingestion
  Sulfur ingestion
  Dilutional acidosis

*RTA,* Renal tubular acidosis.
The items in bold are most common or notable.

Table_15_4

**Table 15.5 Causes of Respiratory Acid-Base Disorders**

**Alkalosis**

Central nervous system stimulation
  Pain
  Anxiety
  Psychosis
  Fever
  Cerebrovascular accident
  Meningitis
  Encephalitis
  Tumor
  Trauma
Hypoxemia or tissue hypoxia
  High-altitude acclimatization
  Pneumonia, pulmonary edema
  Aspiration
  Severe anemia
Drugs or hormones
  Pregnancy (progesterone)
  Salicylates
  Nikethamide
Stimulation of chest receptors
  Hemothorax
  Flail chest
  Cardiac failure
  Pulmonary embolism
Miscellaneous
  Septicemia
  Hepatic failure
  Mechanical hyperventilation
  Heat exposure
  Recovery from metabolic acidosis

**Acidosis**

Central nervous system depression
  Drugs (anesthetics, morphine, sedatives)
  Stroke
  Infection
Airway obstruction
  Asthma
Parenchyma disease
  Emphysema/chronic obstructive pulmonary disease
  Pneumoconiosis
  Bronchitis
  Adult respiratory distress syndrome
  Barotrauma
Mechanical ventilation
  Hypoventilation
  Permissive hypercapnia
Neuromuscular
  Poliomyelitis
  Kyphoscoliosis
  Myasthenia gravis
  Muscular dystrophies
  Multiple sclerosis
Miscellaneous
  Obesity
  Hypoventilation

Table_15_5

**Table 15.6 Differential Diagnosis of Nonanion Gap (Hyperchloremic) Metabolic Acidosis**

GASTROINTESTINAL BICARBONATE LOSS

**Diarrhea**
External pancreatic or small bowel drainage
Uterosigmoidostomy, jejunal loop
Drugs
  Calcium chloride (acidifying agent)
  Magnesium sulfate (diarrhea)
  Cholestyramine (bile acid diarrhea)

RENAL ACIDOSIS

**Hypokalemia**
Proximal RTA (type 2)
Distal (classical) RTA (type 1)
Drug-induced acidosis
  Acetazolamide and topiramate (proximal RTA)
  Amphotericin B (distal RTA)
  Ifosfamide
**Hyperkalemia**
**Generalized distal nephron dysfunction (type 4 RTA)**
  Mineralocorticoid deficiency (e.g., **low renin/aldosterone in diabetic nephropathy**)
  Mineralocorticoid resistance (PHA-1 autosomal dominant)
  Voltage defects (PHA-1, autosomal recessive)
  PHA-2
  ↓ $Na^+$ delivery to distal nephron
  Tubulointerstitial disease
**Drug-induced acidosis**
  Potassium-sparing diuretics (amiloride, triamterene, spironolactone)
  Trimethoprim
  Pentamidine
  Angiotensin-converting enzyme inhibitors
  Angiotensin II receptor blockers
  Nonsteroidal antiinflammatory drugs
  Cyclosporine, tacrolimus
**Normokalemia**
  **Chronic kidney disease (stage 3-4)**
**Other**
Acid loads (ammonium chloride, hyperalimentation)
Loss of potential bicarbonate: ketosis with ketone excretion
Dilution acidosis (rapid saline administration)
Hippurate
Cation exchange resins

*PHA,* Pseudohypoaldosteronism; *RTA,* renal tubular acidosis.
Bold = common in adults.

Table_15_6

**Table 15.7 Proximal Renal Tubular Acidosis**

**Isolated Pure Bicarbonate Wasting (Unassociated With Fanconi Syndrome)**

Primary
Inherited autosomal recessive with ocular abnormalities (missense mutations of *SLC4A4*)
Autosomal dominant with short stature (mutation of *SLC9A3/NHE3*)
Carbonic anhydrase deficiency, inhibition, or alteration
  Drugs: acetazolamide, topiramate, sulfanilamide, mafenide acetate
  Carbonic anhydrase II deficiency with osteopetrosis (mixed proximal and distal RTA –type 3)

**Generalized (Associated With Fanconi Syndrome)**

Primary (without associated systemic disease)
  Genetic
  Sporadic
Genetically transmitted systemic diseases
  Cystinosis
  Lowe syndrome
  Wilson syndrome
  Tyrosinemia
  Galactosemia
  Hereditary fructose intolerance (during fructose ingestion)
  Metachromatic leukodystrophy
  Pyruvate carboxylase deficiency
  Methylmalonic acidemia
Dysproteinemic states
  Multiple myeloma, monoclonal gammopathy
Secondary hyperparathyroidism with chronic hypocalcemia
  Vitamin D deficiency or resistance, vitamin D dependency
Drugs or toxins
  Ifosfamide, lead, outdated tetracycline, 3-methylchromone, streptozotocin, amphotericin B (historic)
Tubulointerstitial diseases
  Sjögren syndrome
  Medullary cystic disease
  Renal transplantation
Other renal and miscellaneous diseases
  Nephrotic syndrome, amyloidosis, paroxysmal nocturnal hemoglobinuria

Table_15_7

**Table 15.8 Genetic and Molecular Bases of Distal Renal Tubular Acidoses (RTA)**

| Clinical Syndrome | Gene/Mutation |
|---|---|
| **Classical Distal RTA** | |
| Autosomal dominant dRTA | *SLC4A1* gene that encodes the AE1 HCO3⁻/Cl⁻ exchanger (band 3 protein). Transporter may be mistargeted to apical membrane |
| Autosomal recessive dRTA with sensorineural deafness | ATP6V1B1 encoding B1-subunit of the apical H+-ATPase in distal tubule<br>Forkhead transcription factor, FOXI1 |
| Autosomal recessive dRTA with normal hearing | *ATP6V0a4* and *ATP6V1C2* encoding the a4 and C subunits of the apical $H^+$-ATPase, respectively<br>*WDR72*, associated with enamel hypoplasia |
| Autosomal recessive mixed distal and proximal RTA with osteosclerosis | *CAII* encoding the carbonic anhydrase type II in red blood cells, bone, kidney |
| Endemic dRTA (northeastern Thailand) | Possible abnormality in $H^+$,$K^+$- ATPase |
| Acquired dRTA | Reduced apical expression of $H^+$-ATPase (Sjögren syndrome) |
| **Generalized Distal Nephron Dysfunction** | |
| Pseudohypoaldosteronism type 1, autosomal recessive | Loss-of-function mutations of ENaC; mutations of genes encoding three subunits of ENaC |
| Pseudohypoaldosteronism type 1, autosomal dominant | Heterozygous mutations of mineralocorticoid receptor gene |
| Pseudohypoaldosteronism type 2 | Mutations in WNK1 and WNK4, and in cullin-RING ligase (CRL) proteins cullin 3 (Cul3) 4 and kelch-like 3 (KLHL3) constitutively activate NCCT, increasing NaCl absorption in distal convoluted tubule |

*ENaC,* Epithelial sodium channel.

Table_15_8

**Table 15.9 Classical Distal Renal Tubular Acidosis**

**Primary**

**Familial**
- Autosomal dominant
  - *AE1* gene
- Autosomal recessive
  - With deafness (*rdRTA1* or *ATP6V1B1* gene)
  - Without deafness (*rdRTA2* or *ATP6V0A4*)
- Sporadic

**Endemic**
- Northeastern Thailand

**Secondary to Systemic Disorders**
- Autoimmune Diseases: Sjögren syndrome: Hyperglobulinemic purpura, cryoglobulinemia, thyroiditis, human immunodeficiency syndrome nephropathy, fibrosing alveolitis, chronic active hepatitis, primary biliary cirrhosis, polyarthritis nodosa
- Hypercalciuria and Nephrocalcinosis: Primary hyperparathyroidism, vitamin D intoxication, hyperthyroidism, idiopathic hypercalciuria, medullary sponge kidney, Wilson disease, Fabry disease, hereditary fructose intolerance, X-linked hypophosphatemia
- Drug- and Toxin-Induced Disease: Amphotericin B, toluene, cyclamate, mercury, hepatic cirrhosis, vanadate, ifosfamide, **lithium,** classical analgesic nephropathy, foscarnet
- Tubulointerstitial Diseases: Balkan nephropathy, kidney transplantation, chronic pyelonephritis, leprosy, obstructive uropathy, jejunoileal bypass with hyperoxaluria, vesicoureteral reflux
- Associated with Genetically Transmitted Diseases: Ehlers-Danlos syndrome, hereditary elliptocytosis, sickle cell anemia, Marfan syndrome, medullary cystic disease, jejunal bypass with hyperoxaluria, hereditary sensorineural deafness, carnitine palmitoyltransferase deficiency, osteopetrosis with carbonic anhydrase II deficiency (mixed proximal-distal RTA type 3)

Table_15_9

**Table 15.10 Forms of Alkali Replacement**

| | |
|---|---|
| Shohl's Solution: $Na^+$ citrate 500 mg, citric acid 334 mg/5 mL | Each 1 mL contains 1 mEq sodium and is equivalent to 1 mEq of bicarbonate |
| $NaHCO_3$ tablets | 3.9 mEq/tablet (325 mg)<br>7.8 mEq/tablet (650 mg) |
| Baking soda | 60 mEq/teaspoon |
| K bicarbonate or citrate | 25–50 mEq/tablet |
| $Na^+$ and $K^+$ citrate solutions<br>$Na^+$ citrate 500 mg, and $K^+$ citrate 550 mg/5 mL or $K^+$ citrate 550 mg, citric acid 334 mg/5 mL | Each 1 mL typically contains 1 mEq potassium and 1 mEq sodium and is equivalent to 2 mEq bicarbonate |
| K citrate / citric acid crystals:<br>$K^+$ citrate 3300 mg, citric acid 1002 mg/packet | Each packet usually contains 30 mEq potassium and is equivalent to 30 mEq bicarbonate |
| $K^+$ citrate tablets | 5 or 10 mEq/tablets |

Doses mentioned are typical but may differ between preparations and must be calculated before prescribing.

Table_15_10

**Table 15.11 Disorders With Dysfunction of Renal Acidification—Generalized Abnormality of Distal Nephron With Hyperkalemia**

**Mineralocorticoid Deficiency**

**Primary Mineralocorticoid Deficiency**
- Combined deficiency of aldosterone, deoxycorticosterone, and cortisol
  - Addison disease, bilateral adrenalectomy, bilateral adrenal destruction, hemorrhage, or carcinoma
- Congenital enzymatic defects
  - 21-hydroxylase deficiency, 3β-hydroxydehydrogenase deficiency, Desmolase deficiency
- Isolated (selective) aldosterone deficiency
  - Chronic idiopathic hypoaldosteronism, heparin (low-molecular-weight or unfractionated) administration in critically ill patient, familial hypoaldosteronism, corticosterone methyl oxidase deficiency types 1 and 2, primary zona glomerulosa defect, transient hypoaldosteronism of infancy, persistent hypotension, and/or hypoxemia

**Secondary Mineralocorticoid Deficiency**
- Angiotensin II–converting enzyme inhibition
  - Endogenous, **angiotensin-converting enzyme inhibitors and angiotensin II receptor antagonists**
- Hyporeninemic hypoaldosteronism
  - **Diabetic nephropathy, tubulointerstitial nephropathies, nephrosclerosis, nonsteroidal antiinflammatory agents**, acquired immunodeficiency syndrome, immunoglobulin M monoclonal gammopathy, **obstructive uropathy**

**Mineralocorticoid Resistance**
- PHA-1—autosomal dominant (human mineralocorticoid receptor defect)

**Renal Tubular Dysfunction (Voltage Defect)**
- PHA-1—autosomal recessive
- PHA-2—autosomal dominant
- Drugs that interfere with $Na^+$ channel function in the CCT
  - **Amiloride, triamterene, trimethoprim**, pentamidine
- Drugs that interfere with $Na^+$-$K^+$-ATPase in the CCT
  - **Cyclosporine, tacrolimus**
- Drugs that inhibit aldosterone effect on the CCT
  - **Spironolactone, epleronone**
- Disorders associated with tubulointerstitial nephritis and renal insufficiency
  - Lupus nephritis, methicillin nephrotoxicity, obstructive nephropathy, kidney transplant rejection, sickle cell disease, Williams syndrome with uric acid nephrolithiasis

Table_15_11

**Table 15.12 Effects of Hyperkalemia on Ammonium Excretion**

- Decrease in $NH_4^+$ production
- Decrease in $NH_4^+$ absorption in thick ascending limb of loop of Henle
- Decrease in interstitial $NH_4^+$ concentration
- Impaired countercurrent multiplication
- Decrease in $NH_3$/ $NH_4^+$ secretion into outer and inner medullary collecting ducts

Table_15_12

**Table 15.13 Hyporeninemic Hypoaldosteronism: Typical Clinical Features**

- Mean age 65 yrs
- Asymptomatic hyperkalemia (75%)
- Hyperchloremic metabolic acidosis (>50%)
- Renal insufficiency (70%)
- Diabetes mellitus (50%)
- Hypertension (75%)
- Weakness (25%)
- Arrhythmia (25%)
- Congestive heart failure (50%)

Table_15_13

**Table 15.14 Treatment of Generalized Dysfunction of the Nephron With Hyperkalemia**

- Loop diuretic (furosemide, bumetanide)
- Alkali therapy (Shohl solution or $NaHCO_3$ tablets)
- $K^+$ binding resin
- Low-potassium diet
- Fludrocortisone (0.1-0.3 mg/day)
  - Avoid in hypertension, volume expansion, heart failure
  - Combine with loop diuretic
- Avoid drugs associated with hyperkalemia, herbs and over-the-counter preparations containing potassium
- In pseudohypoaldosteronism type 1, add NaCl tablets

Table_15_14

**Table 15.15 Contrasting Features and Diagnostic Studies in Renal Tubular Acidosis**

| | Type of Renal Tubular Acidosis | | |
|---|---|---|---|
| **Finding** | **Proximal** | **Classical Distal** | **Generalized Distal Dysfunction** |
| Plasma [$K^+$] | Low | Low | High |
| Urine pH with acidosis | <5.5 | >5.5 | <5.5 or >5.5 |
| Fractional bicarbonate excretion | 10%-15% during alkali therapy | 2%-5% | 5%-10% |
| Associated features | Fanconi syndrome | Nephrocalcinosis/Hyperglobulinemia | Renal insufficiency |

Table_15_15

**Table 15.16 Metabolic Acidosis With High Anion Gap**

**Conditions Associated With Type A Lactic Acidosis**
- Hypovolemic shock
- Cholera
- Septic shock
- Cardiogenic shock
- Regional hypoperfusion
- Severe hypoxia
  - Severe asthma
  - Carbon monoxide poisoning
  - Severe anemia

**Conditions Associated With Type B Lactic Acidosis**
- Liver disease
- Diabetes mellitus
  - **Metformin**

Catecholamine excess

Thiamine deficiency
- Intracellular inorganic phosphate depletion
- Intravenous fructose, xylose, or sorbitol

Alcohols and other ingested compounds metabolized by alcohol dehydrogenase
- Ethanol, methanol, ethylene glycol, propylene glycol

Mitochondrial toxins
- **Salicylates**, cyanide, 2,4-dinitrophenol, **Nonnucleoside anti–reverse transcriptase drugs**

Other drugs (propofol, linezolid)

Metastatic tumors (large tumors with regional hypoxemia or liver metastasis)

**Seizures**

Inborn errors of metabolism

**D-Lactic Acidosis**
- Short bowel syndrome, ischemic bowel, small-bowel obstruction

**Ketoacidosis**
- Diabetic
- Alcoholic

**Intoxications**
- Salicylates
- Ethylene glycol
- Methanol
- Pyroglutamic acid

**Uremia (Late Renal Failure)**

Table_15_16

**Table 15.17 Causes of Metabolic Alkalosis**

**Exogenous $HCO_3^-$ Loads**
- Acute alkali administration
- **Milk-alkali syndrome**

**Effective ECV Contraction**
- Gastrointestinal origin
  - **Vomiting**
  - **Gastric aspiration**
  - Congenital chloridorrhea
  - Villous adenoma
- Renal origin
  - **Diuretics (especially thiazides and loop diuretics)**
  - **Edematous states**
  - Posthypercapnic state
  - Hypercalcemia-hypoparathyroidism
  - Recovery from lactic acidosis or ketoacidosis
  - Nonreabsorbable anions such as penicillin, carbenicillin
  - $Mg^{2+}$ deficiency
  - **$K^+$ depletion**
  - Bartter syndrome (loss-of-function mutations in thick ascending limb of Henle)
  - Gitelman syndrome (loss of function of $Na^+/Cl^-$ cotransporter—DCT)
  - Carbohydrate refeeding after starvation

**Hypermineralocorticoidism**
- Associated with high renin level
  - Renal artery stenosis
  - Accelerated hypertension
  - Renin-secreting tumor
  - Estrogen therapy
- Associated with low-renin level
  - **Primary aldosteronism**
  - Adrenal adenoma
  - Hyperplasia
  - Carcinoma
  - Glucocorticoid suppressible
  - Adrenal enzymatic defects
    - 11β-hydroxylase deficiency
    - 17α-hydroxylase deficiency
  - Cushing syndrome or disease
    - Ectopic corticotrophin
    - Adrenal carcinoma
    - Adrenal adenoma
    - Primary pituitary
  - Other
    - Licorice
    - Carbenoxolone
    - Smokeless (chewing) tobacco
    - Lydia Pinkham tablets

**Gain-of-Function Mutation of ENaC**
- Liddle syndrome

Table_15_17

**Table 15.18 Diagnosis of Metabolic Alkalosis**

| **Saline-Responsive Alkalosis** | **Saline-Unresponsive Alkalosis** |
|---|---|
| **Low Urinary [$Cl^-$] (<10 mEq/L)** | **High or Normal Urinary [$Cl^-$] (>15-20 mEq/L)** |
| Normotensive | Hypertensive |
| Vomiting | Primary aldosteronism |
| Nasogastric aspiration | Cushing's syndrome |
| Diuretic use (distant) | Renal artery stenosis |
| Post-hypercapnia | Renal failure plus alkali therapy |
| Villous adenoma | Liddle syndrome |
| Bicarbonate treatment of organic acidosis | Normotensive |
| $K^+$ deficiency | $Mg^{2+}$ deficiency |
| | Severe $K^+$ deficiency |
| | Bartter syndrome |
| | Gitelman syndrome |
| | Diuretic use (recent) |

Table_15_18